ADVANCED MICROECONOMETRICS
Problem Set - Day 1

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Due date: July 27, 2026

Total points: 100. Please return the problem set on time. Use your favorite software to answer the questions. Submit your problem set to: surzua@umd.edu

Q 1. Consider the following regression model:

$$Y_i = \alpha_0 + \alpha_1 X_i + V_i.$$

Assume the following parameterization of the model: $X \sim N(2, 2)$, $V \sim N(0, 4)$, $\alpha_0 = 2$, and $\alpha_1 = -1$.

(a) Using this parameterization, generate a fake sample of 1,000 observations.
(b) Code the associated likelihood function.
(c) Estimate the parameters of the model using information on $\{Y_i, X_i\}_{i=1}^{1000}$ and your ML routine. Compare your results with the actual values.
(d) Compute standard errors for your point estimates.

Q 2. Consider the following model:

$$D = 1[\alpha_0 + \alpha_1 Z > V].$$

and assume you observe $Y = a + bX + \epsilon$ if $D = 1$ and you observe $Y = 0$ otherwise.

Assume the following parameterization of the model: $Z \sim N(0, 2)$, $X \sim N(0, 1)$, $\epsilon \sim N(0, 1)$, $V \sim N(0, 1)$, $a = 0.5$, $b = -0.5$, $\alpha_0 = 2$, and $\alpha_1 = -1$.

(a) Using this parameterization, generate a fake sample of 1,000 observations.
(b) Code the likelihood function associated with this problem.
(c) Estimate the parameters of the model using information on (Y, D, Z, X) and your ML routine. Compare your results with the actual values.
(d) Compute standard errors for your estimates.

Q 3. Consider the following random utility model:

$$\begin{aligned} Y_1^* &= \beta_1 Z + V_1 \\ Y_2^* &= \beta_2 Z + V_2 \\ Y_3^* &= \beta_3 Z + V_3 \end{aligned}$$

and define $D = 1$ if $(Y_1^* > Y_2^*, Y_1^* > Y_3^*)$,$D = 2$ if $(Y_2^* > Y_1^*, Y_2^* > Y_3^*)$, and $D = 3$ otherwise, with

$$V \sim N\left[\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{11}^2 & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{22}^2 & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_{33}^2 \end{pmatrix}\right].$$

Consider alternative 3 as the benchmark, and assume the following parameterization: $Z \sim N(0, 3)$,

$$\begin{bmatrix} V_1 - V_3 \\ V_2 - V_3 \end{bmatrix} \sim N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.5 \\ 0.5 & 2 \end{pmatrix}\right]$$

$\beta_1 = 0.25$, and $\beta_2 = 0.5$, $\beta_3 = 1$.

(a) Using this parameterization, generate a fake sample of 10,000 observations.

(b) Code the associated likelihood function.

(c) Show that you can identify $\beta_1 - \beta_3$, $\beta_2 - \beta_3$, $Cov\,(V_1 - V_3, V_2 - V_3)$, and $Var\,(V_2 - V_3)$.

(d) Estimate the parameters of the model using information on (D_1, D_2, D_3, Z) and your ML routine. Compare your results with their actual values. (Hint: Don't forget to impose the normalizations).

Q 4. Consider the selection model discussed in Heckman 1974.

(a) Describe the equations linking theory and practice. What are the key assumptions?

(b) Download the NLSY97. Create a data set with the variables utilized by Heckman. Present descriptive statistics for your sample.

(c) Using the NLSY97, estimate the model in Heckman 1974. (Discuss your optimization routine).

(d) Estimate Heckman 1997 for different schooling levels (12 or less years of education versus more than 12 years of education)

(e) Explain under what assumption the average observed wage would be larger than the unconditional average wage.